



Object Oriented Programming using Java Laboratory (DJS23FLES201)

F.Y B. Tech, Semester: II

Experiment list

(AY: 2025-26)

1. To implement Java control statements, loops, and command line arguments (CO1)

- a. Given an integer, n, perform the following conditional actions:
 - If n is odd, print Weird
 - If n is even and in the inclusive range of 2 to 5, print Not Weird
 - If n is even and in the inclusive range of 6 to 20, print Weird
 - If n is even and greater than 20, print Not Weird
- b. WAP to find largest of 3 numbers using nested if else and nested ternary operator.
- c. Write a Java program that reads a positive integer from **command line** and count the number of digits the number (less than ten billion) has.
- d. Write a menu driven program using switch case to perform mathematical operations.
- e. WAP to find grade of student from input marks using if else ladder.
- f. WAP to print the sum of following series $1+1/2^2+1/3^2+1/4^2+\dots+1/n^2$
- g. WAP to display the following patterns:

```
1
2  1
1  2  3
4  3  2  1
1  2  3  4  5
6  5  4  3  2  1
1  2  3  4  5  6  7
```

A

CB

FED

JHGH



2. To implement Arrays (CO1)

- a. You have been given an array of positive integers $A_1, A_2, \dots, A_n$ with length N and you have to print an array of same length (N) where the values in the new array are the sum of every number in the array, except the number at that index.

i/p 1 2 3 4

For the 0th index, the result will be $2+3+4=9$, similarly for the second, third and fourth index the corresponding results will be 8, 7 and 6 respectively.

i/p 4 5 6

o/p 11 10 9

- b. The annual examination results of 5 students are tabulated as follows:

Roll No	Subject1	Subject2	Subject3

WAP to read the data and determine the following

Total marks obtained by each student

The student who obtained the highest total marks

- c. WAP to display following pattern using irregular arrays (jagged arrays).

1

1 2

1 2 3

3. To implement Strings (CO1)

- a. WAP to find out number of uppercase & lowercase characters, blank spaces and digits from the string.
- b. Write a Java program to accept a string and perform the following operations using the **String** class: Find the length of the string, Convert the string to uppercase, concatenate another string to it, Display the original string to show that it is unchanged.
- c. WAP to check if a string is a palindrome or not using inbuilt functions.



- d. Write a Java program to demonstrate the difference between String and StringBuffer by:
Modifying a String using concat() Modifying a StringBuffer using append() Displaying the results
- e. Write a Java program to compare two strings that have the same content but are created in different ways—one using a string literal and the other using the new keyword. Compare the strings using: equals() method to check content equality == operator to check reference equality

4. To implement collections (Array List/ Vectors) (CO1)

- a. WAP to accept students name from command line and store them in vector.
- b. WAP to add n strings in a vector array. Input new string and check if it is present in the vector. If present delete it else add to the vector
- c. A university wants to store and manage the marks of students in a subject using Java Vector collection. Write a Java program that performs the following operations using a Vector<Integer>: Create a Vector to store integer marks. Add marks of 5 students to the Vector. Display all the marks stored in the Vector. Display total number of elements using size(), Capacity of the Vector using capacity(), Insert a new mark at a given index, Remove a mark from a specified index, Check whether a particular mark exists in the Vector, Display the first and last mark in the Vector.

5. To implement class with members and methods (static, non-static, recursive and overloaded methods) (CO1)

- a. Create a class employee with data member's empid, empname, designation and salary. Write methods getemployee() to take user input, showgrade() to display grade of employee based on salary, showemployee() to display details of employee.
- b. WAP to display area of square and rectangle using the concept of **overloaded functions**.
- c. WAP to find value of y using recursive function (static), where $y=x^n$
- d. WAP to count the number of objects made of a particular class using static variable and static method and display the same.



6. To implement array of objects and passing/ returning objects (CO1)

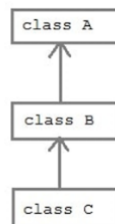
- WOOP to arrange the names of students in descending order of their total marks, input data consists of students details such as names, ID.no, marks of maths, physics, chemistry. (Use array of objects)
- WAP to perform mathematical operations on 2 complex numbers by passing and returning object as argument. Show the use of **this pointer**.

7. To implement Constructors and constructor overloading (CO1)

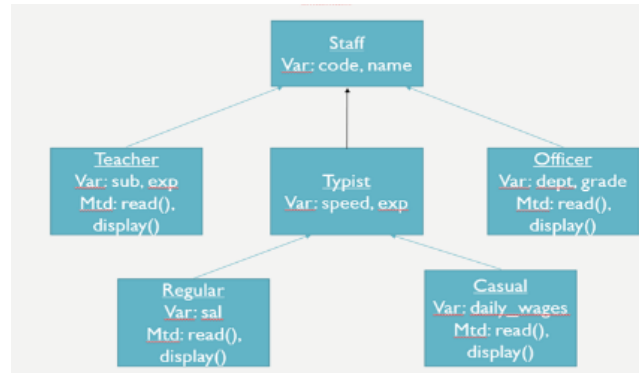
- WAOOP to count the no. of objects created of a class using constructors.
- WAP to display area of square and rectangle using the concept of overloaded constructor (use parameterized, non-parameterized and copy constructor).

8. To implement Inheritance and super keyword (CO1)

- WAP to demonstrate the role of Constructors in inheritance in the following class diagram.

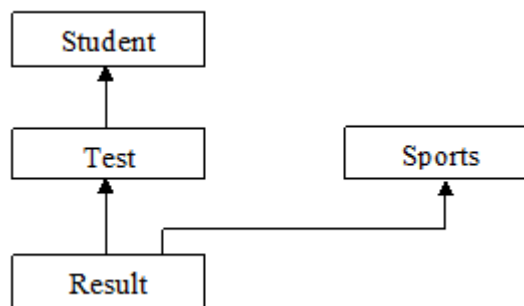


- WAP to create a super class having a variable. Let the variable be initialized to some value within a constructor. This class should have a method display () to display the initial value of the variable. Derive a sub class that accesses the constructor, variable and method of the super class using super keyword.
- Display data of the specialized classes given in the following class diagram.



9. To implement multiple inheritance using interfaces and method overriding (CO1)

- Design an interface with a method reversal. This method takes a string as input and returns the reversed string. Create a class that implements the above interface.
- WAP to implement three classes namely Student, Test and Result. Student class has member as rollno, and read(). Test class has members as sem1_marks and sem2_marks and read(). Result class has member as total. Create an interface named sports that has a member score (). Derive Test class from Student and Result class has multiple inheritances from Test and Sports. Total is formula based on sem1_marks, sem2_mark and score. Use super keyword.



10. To implement dynamic polymorphism, final keyword and garbage collection (CO1)

- Demonstrate using a suitable example that a base class reference variable can point to a child class object or a base class object using the concept of dynamic method dispatch (dynamic polymorphism).
- Adwita, 8th Grade student wants to write a functions to calculate simple interest, compound interest. She wants to keep same (final) rate of interest for every input of principal and time. She wants to ensure that the declared functions are not overridden in any subclasses and the



- class is not inherited by any other class. Help her to declare the variables methods and classes and write the code for the same using final keyword.
- c. Write a Java program to demonstrate **Garbage Collection** by creating an object, making it eligible for garbage collection, and verifying that the object is removed from memory. Hint: Using WeakReference

11. To implement Abstract classes and packages (CO1)

- a. Write an abstract class program to calculate area of circle, rectangle and triangle.
- b. WAP to create a package called vol having Cylinder class and volume (). WAP that imports this package to calculate volume of a Cylinder.

12. To implement exceptions in Java (read input using DataInputStream/ BufferedReader classes) (CO1, CO2)

- a. WAP to implement inbuilt exceptions
- i. Checked exceptions (compile time exceptions)
 - a. ClassNotFoundException
 - b. IOException
 - ii. Unchecked exceptions (run time exceptions)
 - a. NumberFormatException
 - b. ArithmeticException
 - c. ArrayIndexOutOfBoundsException
 - d. NullPointerException
- b. Write a Java Program to Create a User Defined Exception class MarksOutOfBoundsException, If Entered marks of any subject is greater than 100 or less than 0, and then program should create a user defined Exception of type MarksOutOfBoundsException and must have a provision to handle it.
- c. Write a Java program that: Accepts two numbers from the user Performs division inside a try block Handles arithmetic exceptions using catch Uses finally to display a termination message Throws a custom exception if the divisor is zero.



13. To implement Multithreading (CO1, CO2)

- Write a Java program to create two threads: Thread-1 prints numbers from 1 to 5, Thread-2 prints characters from A to E. Both threads should run concurrently.
- Write a Java multithreaded program in which two child threads are created from the same thread class. Each child thread should display the pattern `/*/*/*` on the console.
- Write a Java multithreaded program in which the parent (main) thread creates a child thread to calculate the sum of first n natural numbers. The parent thread must wait until the child thread finishes calculation, and then display the result calculated by the child thread.
- Write a Java program in which **two threads share a single bank account object**. Each thread attempts to withdraw money from the account. Use **synchronization** to ensure that the account balance does not become inconsistent.
- setname getname
- Write a Java program in which one thread waits using `wait()` and another thread prints numbers from 1 to 5 and then notifies the waiting thread using `notify()`, to demonstrate inter-thread communication.
- Write a Java program to create two threads, use `setname` to name each of them, then use `getname` to print their names five times. Set different priorities for the threads using `setPriority()`.

14. To implement basic Swing programs with event handling (CO1, CO3)

- Write java program to create a registration form. Take Login id and Password from the user and display it on the third Text Field which appears only on clicking OK button and clear both the Text Fields on clicking RESET button.

Login		_[]X	
Login :		Password :	
		OK	RESET

- Write a program to create a basic calculator.
- Display the selected fields in Details after submit button is clicked



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Welcome to fahmidasclassroom

Name:

Gender: ☐ Male ☐ Female

Interest: ☐ Music ☐ Swimming

Favourite Place:

Details: